



DEVOPS,AWS WITH AI MASTER PROGRAM SYLLABUS

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Technologies

DevOps Master Program

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Note:

- 1) All the topics which we mentioned below, we are going to give training on from basic to advanced level with Hands on Training by DevOps Experts.
- 2) For Each Topic Soft copy material and Interview Questions will provide.
- 3) Real time Use Case Scenarios
- 4) Resume Preparation
- 5) Job Oriented Training

1) AI Topics : The course also introduces AI-powered tools such as

- GitHub Copilot
- Chat GPT: AI-Driven Deep Search, Image Creation, Thinking Tool
- Grok: AI-Driven Presentation & Documentation Tool
- Gamma : AI-Driven Presentation & Documentation Tool
- Napkin: AI-Driven turns your text into visuals so sharing your ideas is quick and effective.
- Merlin: AI-Driven to research and chat with websites, videos, and documents, plus write social media content in 128 languages Tool
- Notebooklm: AI-Driven NotebookLM is a research and note-taking online tool developed by Google Labs
- Julius: AI-Driven data Analysis Tool
- Genspark: AI-Driven Presentation & Documentation , Designer Tool
- PromptOps: It's a Process to how to prompt the question in AI tools.
- Why AI is a Game-Changer for DevOps
- Top AI Tools Accelerating DevOps in 2025 and 2026
- How DevOpsGPT Works: From Idea to Code in Minutes
- Introducing DevOpsGPT: Autonomous AI Agents for Software Creation
- Real-World Impact: AI Tools Driving DevOps Efficiency
- The Future of DevOps is AI-Enabled

And many more things we will cover in our course, helping you automate faster, code smarter, and build intelligent DevOps pipelines.

2) DevOps Introduction

- Software Development Life Cycles (SDLC)
 - Waterfall Model
- Agile Methodology
 - Scrum Model
 - Sprint/ Iteration
- What is DevOps?
- Why DevOps?
- DevOps Importance
- DevOps Model

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- DevOps Life cycle
- Market Trend and Career Scope for DevOps
- DevOps Tools

3) Linux

- Introduction of Linux Operation System
- Installing Pre-requisite Software's (SSH Tools and FTP Tools) in Desktop/Laptop..
- Linux File/Directory structure.
- Create an account in AWS.
- Create an EC2 Instance in AWS
- **Linux Commands**

mkdir	head	uptime	crontab	wget
ls	tail	last	ssh	netstat
tree	more	ps	scp	exit/logout/ctrl+d
cd	less	kill	ssh-keygen	yum
pwd	sort	top	ssh-copy-id	read
rmdir	tr	zip	awk	restart
touch	sed	unzip	cut	reboot
find	grep	tar	free	shutdown
umask	who	useradd	dmidecode	
chmod	w	passwd	mail	
chown	whoami	chage	clear	
chgrp	whereis	groupadd	cal	
cp	date	usermod	tee	
mv	df	id	script	
rm	du	lfd	ping	
wc	hostname	su	telnet	
ln	ipconfig	sudo	history	
vim	man	users	uname	
nano	help	groups	cat /etc/*releases	
echo	info	userdel	watch	
cat	service	groupdel	curl	

4) Shell Scripting

- Introduction
- What is Shell?
- Types of Shells
- What is Shell Scripting?

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- First Shell Script program
- File Naming Conventions
- Comments
- Variables
- Command line Arguments
- String
- Arithmetic Operations
- User Interaction using read command
- Input and Output Redirection
- Control commands – if
- Control commands – for
- Control commands – while loop
- Control commands – Switch case
- Functions
- Pipe

5) Git

- Account Creation in GitHub/GitLab/Bitbucket
- What is git?
- What is the VCS?
- What is SCM?
- What is Branch?
- What is Tag?
- Git Administration.
- Git commands
- Working with git as a Developer perspective
- SSH Key generation
- PAT creation
- Cloning Repositories
- Merging Branches
- Pull Requests
- Cloning Remote Repo
- GitHub API – Some Realtime Shell Script Examples covers
- Forking Repo
- Branching strategy
- README file
- Best practices for Releases/Code commits in any VCS

6) Maven

- Introduction
- Features & Benefits of Maven

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- Installation (Maven Environment Setup)
- Directory Structure
- Content of pom.xml
- Maven Repositories
- Maven Life Cycles
- Executing some Examples

7) Tomcat

- Introduction
- Difference between App server and Web server
- Understanding of Web Servers, App Servers and Database Servers
- Installation (In Windows - Extraction, In Linux Installation)
- Directory structure
- Start the Tomcat server
- Stop the server
- Users creation
- Roles
- Port number change
- Application Deployment
 - Through Admin Console
 - Copy artifact into webapps folder

8) Apache HTTP Server / Nginx Server

- Introduction
- Installation
- Directory Structure
- Start the HTTP server
- Deploy the web static application.

9) SonarQube

- Introduction
- Pre-Requisites
- Architecture
- Installation
- Change the Port Number
- Execution
- Administration
 - Users Creation (Normal User and Administrator)
 - Project Creation
 - Project deletion
 - Token Generation
 - Create Quality Profiles

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- Create Quality Gates
- Configure Email settings
- Sonar Cloud

10) Nexus

- Introduction
- Installation
 - Password and Email change for Admin User
 - Email server Configuration
 - Port Number Change
 - Context root change
- Nexus Directory structure
- Create the Repositories (maven2hosted, maven2proxy, maven2group and docker..)
- Integrate the Maven with Nexus
- Create Clean up Policies and attach policies to Repositories.
- Create Users, Roles
- Nexus API

11) Jenkins

- **Introduction**
 - Continuous Integration (CI)
 - Continuous Delivery (CD)
 - Continuous Deployment (CD)
- **Installation**
 - In Linux Server
- **Create the Maven Project using Freestyle Project type**
 - Integrate Maven software if not done.
 - Integrate Nexus with Jenkins
 - Integrate SonarQube with Jenkins
 - Deploy the App into Tomcat
 - Through “Deploy to container” plugin
 - Through Script – SSH Agent Plugin
 - Configure Email Functionality
 - Poll SCM
 - Build Periodically
 - Git Web Hooks
 - Discard Old Build
 - Disable this project
 - Delete workspace before build starts
 - Add timestamps to the Console Output

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- JACOCO plugin
- **Jenkins Directory structure**
- **Create the Maven Project using Maven Project type**
- **Plugin Management**
 - Safe Restart
 - Next Build Number
 - Email Extension
 - SonarQube Scanner
 - Schedule Build
 - Artifactory Plugin
 - Cloud Foundry
 - Blue Ocean
 - Deploy to container
 - Deploy WebLogic
 - WebSphere Deployer
 - Maven Integration
 - JaCoCo
 - SSH Agent
 - Publish Over SSH
 - Thin Backup
 - Build Name Setter
 - Convert To Pipeline
 - Role-based Authorization Strategy
 - Rebuild last
 - Office 365 connector
- **External Plugins Installation**
 - Urban Code Deploy
- **Port Number Change**
- **Build with parameters**
- **Create View**
- **Jenkins Security**
 - Create Users (Default Admin)
 - Provide the specific access Jenkins
 - Provide the access to specific access to specific projects
- **Upstream and Downstream Jobs**

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- Create the Pipeline Project Jobs (Scripted way and Declarative way)
- Create the Multibranch Pipeline Project Jobs
- CI/CD Implementation for Node JS Project
- Create Master/Slave
- Jenkins Backup
- Jenkins Migration
- Jenkins Shared Libraries
- Slack Notifications
- Some more examples on Declarative Pipeline Script
- Jenkins CLI

**** 5 Ways of Deploying application into Production Environment – Realtime Projects Experience ****

12) Ansible

- Introduction
- What is Ansible?
- Configuration Management
- Ansible Features
- Ansible Architecture
- Host Inventory file
- Ansible Installation
- Ansible Commands
- Develop the Ansible Play Books
- Tags
- Handlers
- Group Variables and Host Variables
- Loops
- Conditional Statements
 - o when condition
- Ansible Vault
- Ansible Roles

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- Working With Dynamic Inventory

13) Resume Preparation and Agile Topics

- Resume Preparation
- DevOps Engineer Roles and Responsibilities
- DevOps Engineer Daily Activities
- Process flow of Scrum Methodologies and Agile Methodology
- Scrum Meeting
- Sprint Planning
- Retrospective Meeting
- Release Management

14) AWS

a) Introduction to Cloud & DevOps

- Role of DevOps Engineer and Platform Engineer.
- Importance of Jenkins, Docker, Kubernetes, AWS and Terraform in DevOps.
- Overview of Cloud, Cloud Computing, Cloud Services and Cloud Providers.
- Why Cloud Adoption...??
- Popular Cloud Providers and AWS Free Tier.

b) Cloud Infrastructure Foundations

- Traditional On-Prem Data Centers.
- Application Deployment Workflow in On-Premise.
- Why Some Companies Still Use On-Prem.
- Traditional Infrastructure Challenges.
- On-Prem vs Cloud [Advantages and Drawbacks].
- Business Requirements Driving Cloud Adoption.

c) AWS Global Infrastructure

- AWS Regions, Availability Zones and Data Centers.
- Why Regions and Availability Zones Matter.
- Choosing the Right Region.
- Enabling Regions.
- Service Quotas.
- Billing and Cost Management.
- AWS Account Setup and Free Tier.
-

d) EC2 [Elastic Compute Cloud]

- EC2 Basics.

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- EC2 Pricing Models [On-Demand, Spot and Reserved].
- How EC2 Billing Works.
- Components of EC2 [AMIs, Instance Types, Tags, Key Pairs, Storage and Security Groups].
- Private IP, Public IP and Elastic IPs.
- EC2 Migration Across Regions Through Amazon Machine Image.

e) Storages

- EBS [Elastic Block Storage]
- Volumes, Types and Snapshots.
- Recover Access When PEM File is Lost.

f) EFS [Elastic File System].

g) S3 [Simple Storage Service]

- Access Control Lists and Bucket Policies.
- Versioning.
- Storage Classes.
- Lifecycle Rules.
- Replication.
- Host a Static Website on S3.

h) VPC [Virtual Private Cloud]

- CIDR, Subnet Masking and Subnets.
- Route Tables.
- Internet Gateway and NAT Gateway.
- Security Groups and NACLs.
- VPC Peering.
- Troubleshooting Network Issues.

i) Load Balancing and DNS [Domain Name System]

- Elastic Load Balancing
- Network Load Balancer [NLB].
- Application Load Balancer [ALB].
- Listeners, Targets and Target Groups.
- Amazon Route 53.
- Domain Purchase.
- Hosted Zones and Name Servers.
- Creating and Configuring DNS Records.
- HTTP Status Codes.
- Map Load Balancer With Domain [Route53].

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j) Auto Scaling

- Auto Scaling Groups
- Launch Templates.
- Scaling Policies.
- End to End Auto Scaling Setup With Application Load Balancer.

k) IAM [Identity and Access Management]

- Users, Groups and Roles.
- Policies.
- Identity Providers.
- Security Best Practices.

l) AWS Concepts for DevOps

- AWS Command Line Interface.
- AWS Secrets Manager.
- Elastic Container Registry [ECR].
- Elastic Kubernetes Service [EKS].
- AWS Cost Optimization.

m) AWS Services vs Azure Services comparison.

15) Docker

a) Docker Foundations

- Evolution of Application Deployment.
- Physical Servers, Virtual Machines and Containers.
- Microservices and Containerization.
- Docker as Cross Platform Container Engine.
- Docker Editions.
- Docker Architecture.

b) Docker Concepts

- Dockerfile.
- Docker Image.
- Docker Container.
- Advantages of Docker Containers.
- Docker Registry and Repositories [Public and Private].

c) Docker Installation

- Install Docker on AWS Ubuntu.
- Install Using User Data.
- Install On Amazon Linux.
- Docker Working Directory Configuration.

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d) Docker Commands

- General Docker Commands.
- Docker Image Commands
- Image Tags and Naming Conventions.
- Custom Dockerfile Names.
- Build Cache and Cache Optimization.
- Inspect Images.
- View Image Layers.
- Dangling Images and Cleanup.
- Scan Docker Images.
- Build Images with Multiple Versions.
- Docker Container Commands
- Logs, Stop, Kill and Prune.
- Copy Images Across Servers Without Registry.
- Non Root User Containers.
- Intermediate Containers.

e) Docker Registry Operations

- Setup DockerHub Account.
- Authenticate With DockerHub.
- Authenticate With AWS ECR.
- Authenticate With Nexus Repository.
- Pull from Public Registry and Push to Organization Registry.

f) Dockerfile

- Dockerfile Instructions.
- Multi Stage Dockerfiles.
- dockerignore File.
- Best Practices for Building Optimized Images.
- Dockerfiles for Microservices.

g) Realtime Git and Docker Integration

- Git Branching Strategies.
- Git Branching Policies.
- Docker Build and Deployment Pipeline Using Jenkins.
- GitHub Webhooks CICD Pipeline.

16) Kubernetes

a) Kubernetes Introduction

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- Kubernetes and Why It Matters.
- Docker and ContainerD.
- Kubernetes Architecture.
- Cluster Components.
- Cluster Capacity Planning.

b) Kubernetes Cluster Setup

- Self Managed Clusters
- kubeadm Introduction.
- RAFT Consensus Algorithm for Multi Master.
- CRI, CNI and CSI.
- Kubernetes Installation using kubeadm.
- Configure Kubectl and Kubeconfig.
- Install Specific kubectl Versions.
- Setup Client Machine for Kubectl.
- Managed Kubernetes Cluster
- Provision Elastic Kubernetes Service.
- Node Groups.
- Elastic Kubernetes Service Deployment Workflow.

c) Kubernetes YAML and Objects

- YAML Structure.
- Different Ways to Create Resources.
- Namespaces.
- PODs [Single and Multi Container].
- Static PODs.

d) Kubernetes Services

- ClusterIP.
- NodePort.
- LoadBalancer.
- Headless Service.
- Image Pull Policies.

e) Labels and Selectors

- Set Based Selectors.
- Equality Based Selectors.

f) Kubernetes Workloads

- Controllers
- ReplicationController.
- ReplicaSet.

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- DaemonSet.
- Deployments.
- Recreate Strategy.
- Rolling Updates.

g) Deployment Strategies

- Canary.
- Blue Green.

h) Resource Management

- Resource Requests and Limits.
- Horizontal POD Autoscaler.

i) Configuration and Secrets

- ConfigMap.
- Secrets.
- Security Context.
- Service Account.
- Pull Images from Private Docker Registry.

j) Kubernetes Storage

- Volumes
- emptyDir.
- hostPath.
- NFS Server.
- Persistent Storage
- Persistent Volume.
- Persistent Volume Claim.
- Static Provisioning
- HostPath Persistent Volume.
- NFS Persistent Volume.
- Dynamic Provisioning
- NFS Provisioner using StorageClass.

k) Application Deployment

- Deploy Spring Boot Application With Environment Variables.
- Liveness and Readiness Probes.
- Stateless vs Stateful Applications.
- StatefulSets.

l) Kubernetes Scheduling

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- Node Name.
- Node Selector.
- Node Affinity.
- Pod Affinity.
- POD Anti Affinity.
- Taints and Tolerations.

m) Node Maintenance

- Cordon, Drain and Uncordon.
- POD Disruption Budgets.
- ResourceQuotas and LimitRanges.

n) Kubernetes Lifecycle Management

- Imperative Commands
- Cluster Upgrade [kubeadm].
- ETCD Backup and Restore.
- Backup Kubernetes Resources.
- Troubleshooting Cluster Issues.

o) Networking and Ingress

- Ingress and Ingress Controllers.
- Gateway API and HTTPRoute.
- Network Policies.

p) Advanced Kubernetes Tooling

- Kubernetes Lens.
- RBAC [Roles, RoleBindings, ClusterRoles and ClusterRoleBindings].
- Helm.
- GitOps with ArgoCD and ApplicationSet.
- KubeContext.
- Kubectl, Kubens and Kubectl Aliases.
- PriorityClass.

q) EKS Advanced Topics

- EKS Cluster Autoscaler.
- EKS Upgrade [AWS CLI and Console].
- Map Load Balancer with Route 53.

r) Monitoring and Logging

- Prometheus and Grafana.
- Elasticsearch, Filebeat and Kibana [EFK].

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s) **CICD for Kubernetes**

- Deploy Microservices on EKS using Jenkins.

t) **Certifications**

- AWS Cloud Practitioner.
- Kubernetes and Cloud Native Associate [KCNA].
- Certified Kubernetes Administrator [CKA].
- Certified Kubernetes Application Developer [CKAD].

u) **Final Deliverables**

- Hands On Labs for Every Concept.
- Live Projects for AWS, Docker and Kubernetes.
- CICD EKS Deployment Project.
- Certification Ready Practice.
- DevOps Best Practices.
- Cloud Architecture Design Skills.

v) **Monitoring Tools (New Relic/Grafana, Prometheus)**

17) **DevOps Master Program Highlights**

- Each Topic Soft Copy Material
- Each Topic Interview Questions
- Real Time Use Case Scenarios
- Case Studies
- Job Oriented Training

Common Questions?

Do We

- **Provide Lifetime Access to DevOps Master Program DevOps tools Documents? Yes**
- **Provide Lifetime Access to Recorded Videos? No, 90 Days**
- **Give Downloaded Access to recorded videos? No**
- **Provide Trainings to 1 to 1? No**
- **Provide Interview Questions for each topic? Yes**
- **Is there any Fast Track? – Contact to trainer for this requirement.**
- **Fee refundable or Transferable? No**
- **Is there any Job Support? – If any openings we will refer to the companies.**

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